

JEREMIAH ADEBAYO

Agent Infrastructure & Machine Learning Engineer

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SUMMARY

IT student and ML/software engineer building autonomous agent systems and production-grade ML pipelines from first principles — a multi-agent SWE automation system (LangGraph, hybrid retrieval, AST-based patching) and a from-scratch anomaly detection pipeline taken through quantization and edge deployment. Seeking remote roles at the intersection of agent infrastructure and applied ML.

PROJECTS

Noctis — Autonomous Multi-Agent Software Engineering System | May – June 2026

- Architected a multi-agent code-fixing pipeline in LangGraph using Send-based parallel fan-out, extending the system from single-file to concurrent multi-file bug resolution across a planner → engineer → critic loop with selective retry routing
- Built a hybrid retrieval layer combining SQLite FTS5/BM25 lexical search with Voyage AI code embeddings (1536-dim) over Qdrant Cloud, feeding relevant repository context to a DeepSeek-based engineering agent
- Implemented AST-level code patching via libcst (vs. naive search/replace) for structurally reliable multi-file edits, orchestrated through a self-hosted LiteLLM proxy for multi-provider model routing (Cerebras, DeepSeek)
- Deployed and containerized the full pipeline (Docker sandboxed test execution) on cloud infrastructure

Nightfall — Multi-Category Visual Anomaly Detection (PatchCore, from scratch) | March – May 2026

- Reimplemented PatchCore (Roth et al., 2021) from the algorithm level — WideResNet50 multi-scale feature hooks, greedy k-center coreset selection, kNN memory-bank scoring — across all 15 MVTec AD categories, achieving 0.938 mean image AUROC with full per-category AUROC/PRO evaluation
- Diagnosed a systemic INT8 quantization failure causing severe accuracy degradation across 8 MVTec categories (AUROC drops of 5–47 points, e.g. hazelnut 0.956→0.486, metal_nut 0.980→0.623); isolated two distinct root causes — confidence-reweighting breakdown and train/test feature-space mismatch — and resolved both via a config-gated fix and consistent-extractor memory bank refit, recovering to 0.894–1.000 AUROC on validated categories
- Built and validated a full deployment pipeline — ONNX export (<1e-5 numerical deviation from native PyTorch), INT8 quantization (3.98x model size reduction), gRPC/REST serving — with real end-to-end inference validated against a Wokwi-simulated ESP32 client over live network

EXPERIENCE

Software Engineering Intern | UniK Connect

March 2026 – Present

- Developed backend functionality in PHP; configured and maintained servers to support networking infrastructure for SME clients

Research & Data Analyst (Volunteer) | OpenGov Africa

August 2025 – Present

- Research role at OpenGov Africa, working independently within a self-directed research team; led data collection and Python/pandas analysis on Freedom of Information (FOI) law implementation across African countries, delivering a full report of findings

TECHNICAL SKILLS

Languages: Python, R, C++, JavaScript, SQL, PHP

ML/AI: PyTorch, Tensorflow, ONNX Runtime, LangGraph, XGBoost, Vector Search (Qdrant), Voyage AI Embeddings

Infrastructure: Docker, gRPC, FastAPI, LiteLLM, SQLite/FTS5, Oracle Cloud(self-managed cloud instances)

Other: Git, REST API design, AST-based code manipulation (libcst), embedded systems (ESP32)

EDUCATION

University of Ilorin (UNILORIN), Nigeria

Expected 2027

B.Sc. Information Technology, Third Year